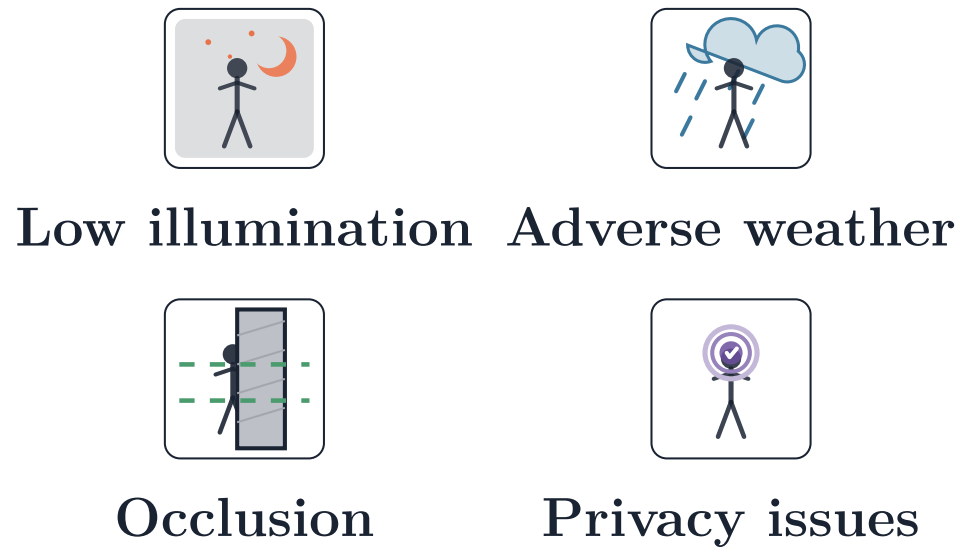


Few-Shot Radar Person Identification via Tracking-Integrated InSAR and Pre-Trained Meta-Learning

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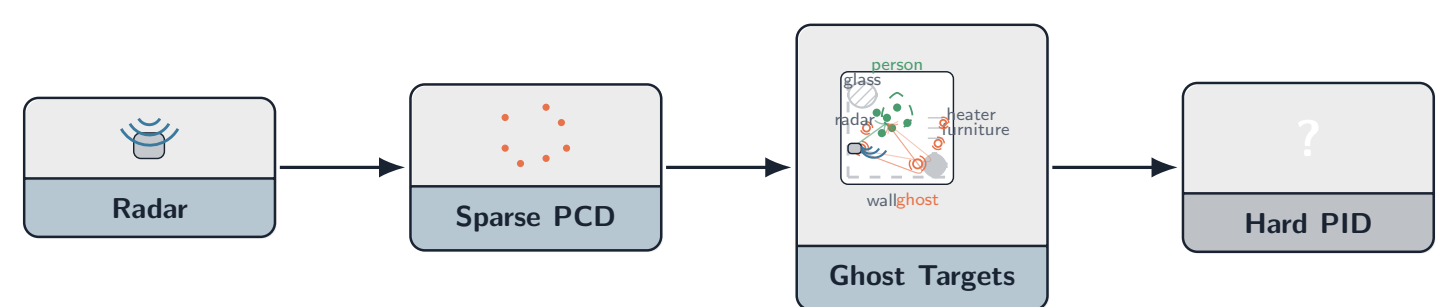
1 Introduction

Vision-based person identification (PID) struggle under conditions [1, 2]:



Radar Advantages: Zero optical exposure · All-weather operation · Penetrates thin barriers · Privacy preserving

2 Challenges in Radar-Based PID



Radar point clouds are sparse, noisy, and contaminated by multipath ghosts



Low Angular Resolution

Conventional angle-of-arrival [3] yields motion-defocused clouds and multipath artifacts.



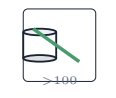
Rigid-Target Bias

Interferometric Inverse Synthetic Aperture Radar (InSAR) improves resolution but is validated only on large *rigid* targets [4, 5].



Multipath Ghosts

Ghosts overlap in range-Doppler and mimic real 3D clusters [6]



Data Scarcity

Existing SOTA [7, 8] demand > 100 seq/person; Point Cloud Transformer (PCT) [9] handles spatial structure but targets *static* objects [10].

Gap: Tracking-integrated InSAR + pre-trained meta-learning.

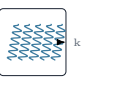
3 Key Results

73.8% InSAR IoU

< 5.5 cm Height Error

70.0% Cross-Session PID

Contributions & System Impact



Articulated Validation: First InSAR validated on humans in multipath-rich indoor envs.



Resolution-level Error: < 5.5 cm (med. 3.87 cm) approaches $\Delta r=4.55$ cm limit (vs. prior 0.2–1.6 m errors).



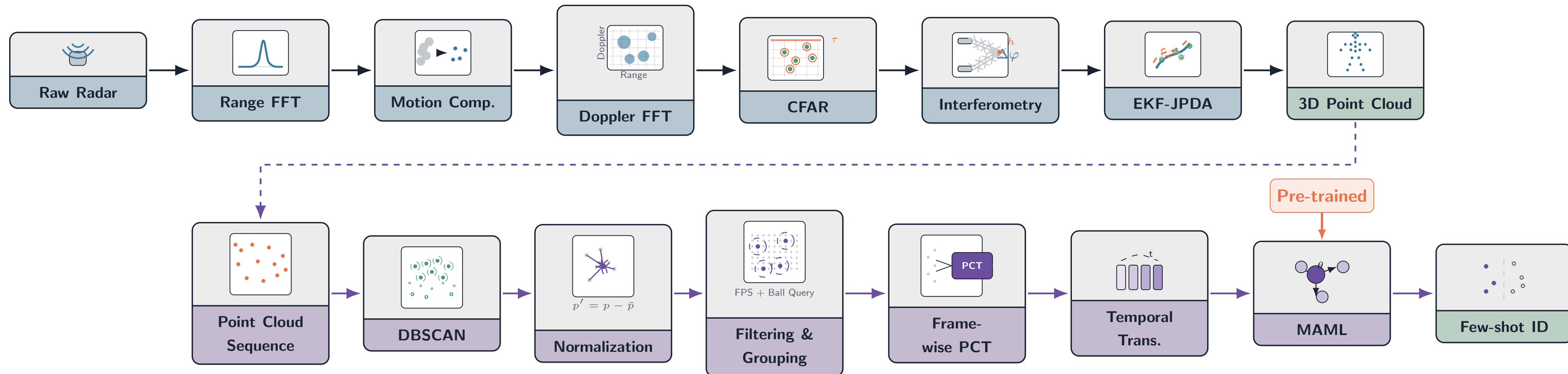
SOTA Supervised: PCT-Trans achieves 98.99% supervised (+2.81pp over TCPCN).



MAML Gains: Pre-trained MAML yields 70.0% cross-session (+15.6pp over random).

Deployable few-shot radar PID with minimal training data.

4 Our Framework: Unified Sensing + Meta-Learning



End-to-end system: InSAR 3D reconstruction $\rightarrow$ spatio-temporal learning $\rightarrow$ PID

(1) Radar domain

Tracking-integrated InSAR reconstructs high-res 3D point clouds of articulated human motion, validated against motion-capture ground truth.

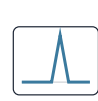
(2) ML domain

Spatio-temporal transformer (PCTrans) embedded in Model-Agnostic Meta-Learning (MAML); task-specific pre-trained initialization substantially improves few-shot adaptation.

(3) System integration

High-quality 3D reconstruction + meta-learning together enable practical few-shot radar PID, reducing dependence on large subject-specific datasets.

5 InSAR Pipeline: 3D Reconstruction



1. Range FFT

- FMCW chirp compression to $S_{R,m}(f, t)$ per m -th antenna.
- Physical range limit $\Delta r = c/(2B)$ bounds our achieved height error.



2. Motion Compensation (ICBA)

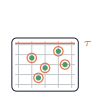
- Translational model $R_0(t) = r_0 + \beta t + \gamma t^2$.
- Nelder-Mead optimizes contrast $[\hat{\beta}, \hat{\gamma}]$ to remove smearing $\rightarrow$ yields *signal-to-noise ratio (SNR) enhanced* $S_{C,m}(f, t)$:

$$[\hat{\beta}, \hat{\gamma}] = \arg \max \left\{ \frac{\sqrt{\frac{1}{N} \sum (I_i - \bar{I})^2}}{I} \right\}$$



3. Coherent Combination & Doppler FFT

- Geometric phase corrections align all MIMO virtual antennas (SNR boost) $\rightarrow$ *focused* 2D Range-Doppler Profile (RDP).



4. CFAR Segmentation

- SOCA-CFAR ($N_T = 6$, $P_{FA} = 10^{-10}$) isolates *target cells* from noise/clutter.

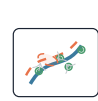


5. Interferometry

- Phase difference between top/bottom MIMO rows *yields elevation*:

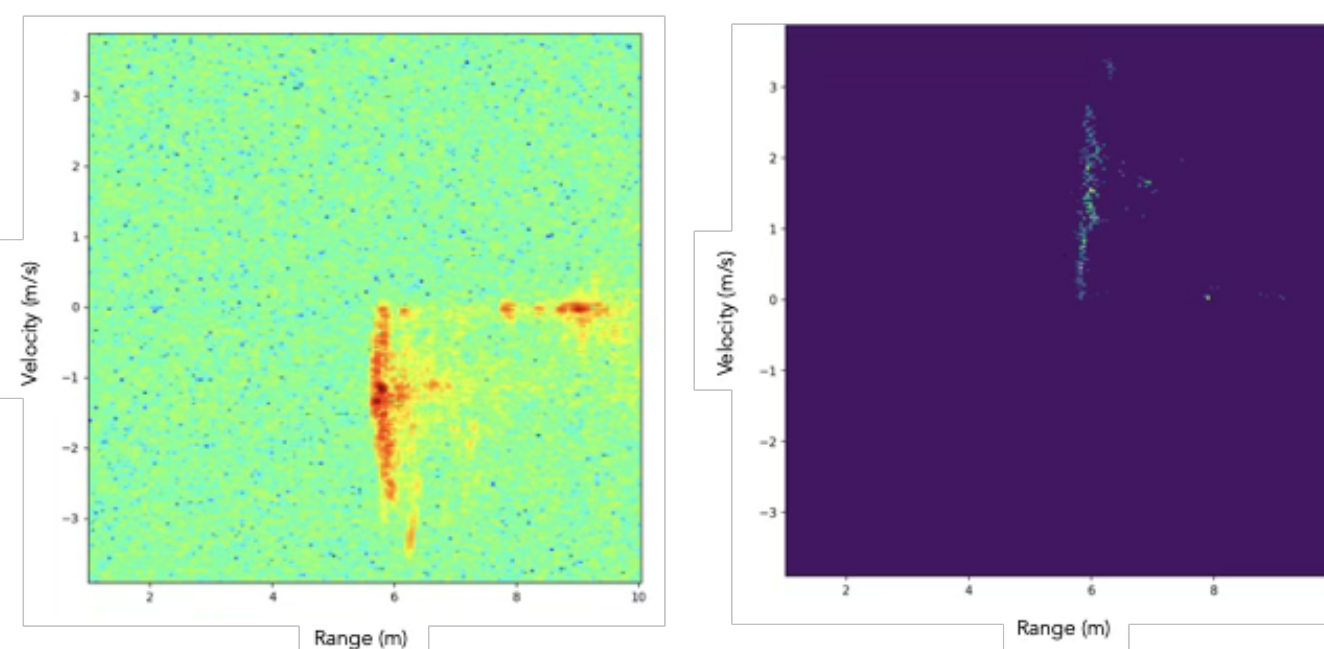
$$I_1 \cdot I_2^* = |I_1| |I_2| \exp(j \frac{2\pi}{\lambda} \Delta R) \rightarrow h = \frac{\lambda \phi R}{2\pi L}$$

- Exploiting MIMO antenna pairs reduces measurement noise.



6. EKF-JPDA Tracking

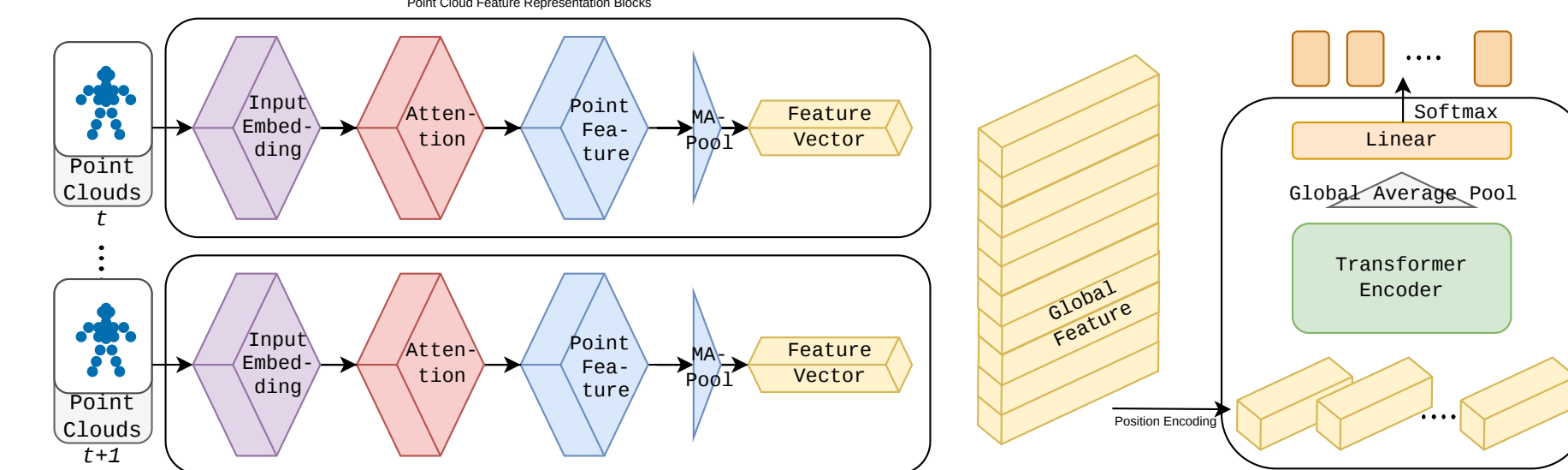
- Spatial-domain JPDA handles data association; EKF filters state covariance.
- Max miss count=1 $\rightarrow$ *prunes multipath ghosts* while tolerating occlusions.



Raw RDP

Post-InSAR RDP

6 Few-Shot PID with PCTrans



PCTrans: Shared Spatial Extractor + Temporal Gait Model



Gait-Invariant Processing

- Centroid norm. $p'_i = p_i - (\bar{x}, \bar{y}, 0)$ prevents path overfitting



Spatial PCT Encoder

- Farthest Point Sampling $\rightarrow$ non-uniform density
- Ball Query $\rightarrow$ rotation-invariant neighborhoods.
- Neighbor Embedding: $\Delta F(p) = \text{concat}_{q \in \text{knn}(p,k)} (F(q) - F(p))$
- Offset-Attention acts as a discrete Laplacian: $F_{\text{out}} = \text{LBR}(F_{\text{in}} - F_{\text{sa}}) + F_{\text{in}}$ ($\mathbf{f}_t \in \mathbb{R}^{192}$)



Temporal Transformer

- Positional encoding injects order
- Multi-head self-attention captures cross-frame gait dynamics: $p(y|\mathcal{S}) = \text{Softmax}(W_2 \cdot \text{ReLU}(W_1 \mathbf{g} + b_1) + b_2)$

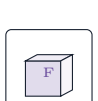
7 Pre-trained Meta-Learning [11]



Two-Phase Domain-Aware Adaptation

- Phase 1:** Supervised pre-training (3 subjects) $\rightarrow \theta_0^{PT}$ (session-invariant patterns).
- Phase 2:** MAML meta-training [12, 13].

Inner: $\theta'_i = \theta - \alpha \nabla \mathcal{L}$ Outer: $\theta \leftarrow \theta - \beta \nabla_{\theta} \mathcal{L}$

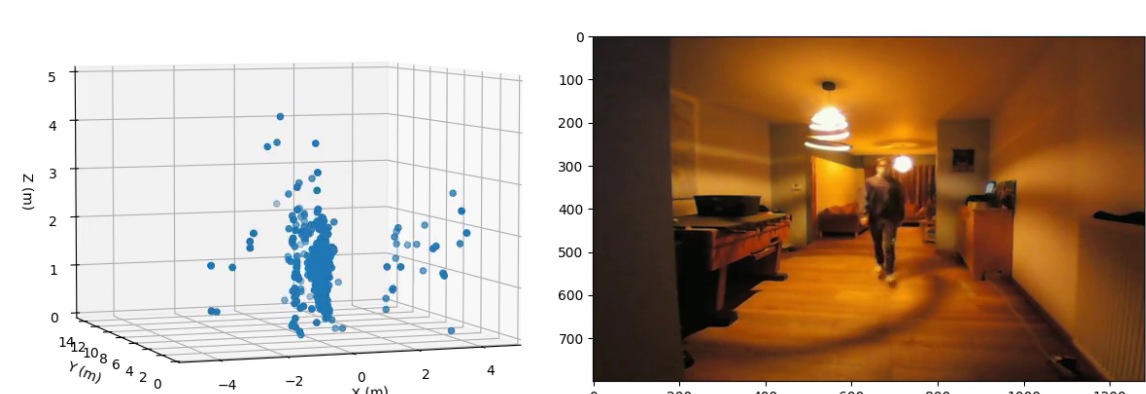


Key Insight: θ_0^{PT} lies closer to optimal θ^* , enabling faster convergence than random initialization.

Gain: +15.6pp over random MAML

8 Experimental Results

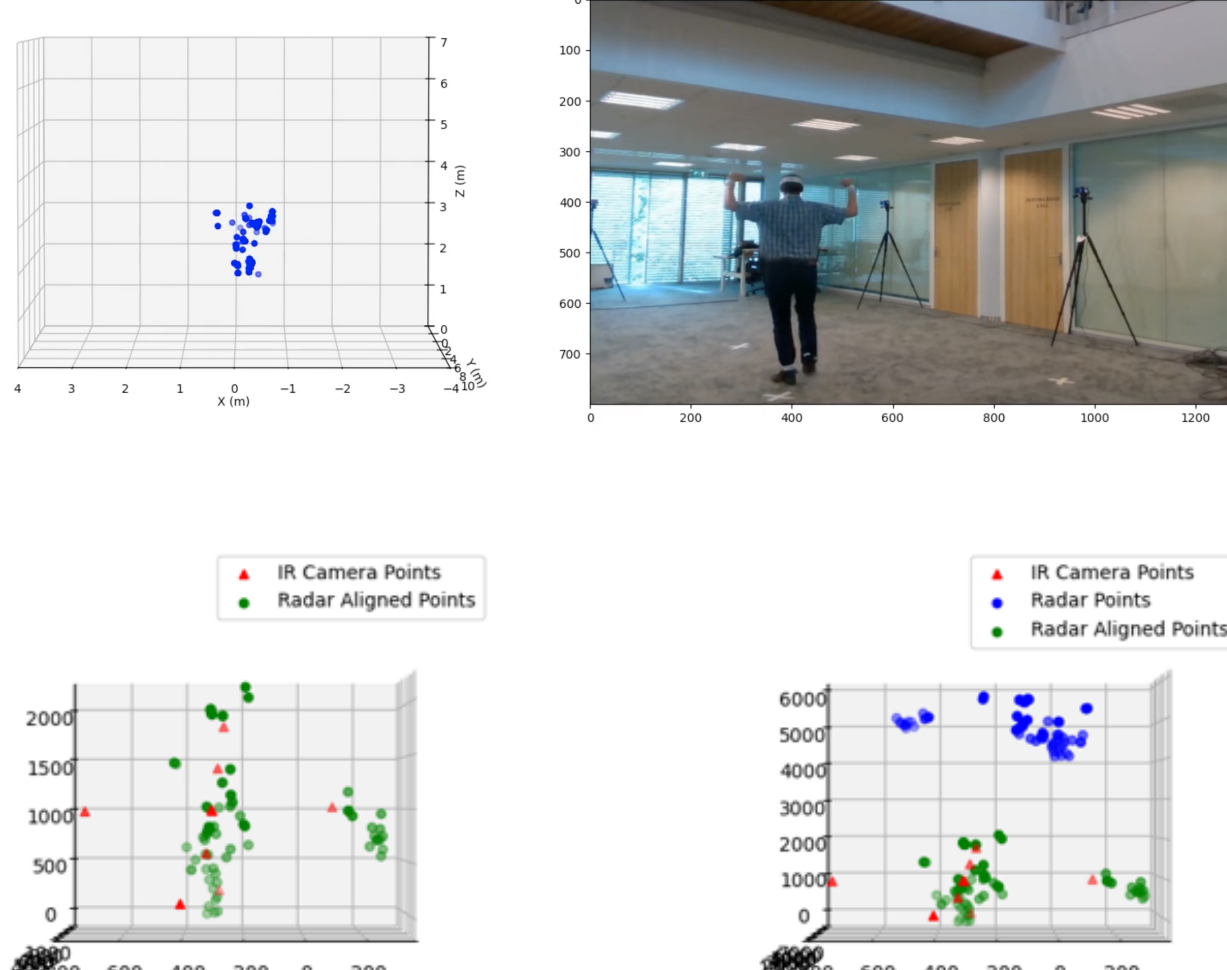
Ghost Removal (EKF-JPDA)



(a) Before: Ghost remains

(b) After: Ghost rejected

InSAR vs Motion Capture



Reconstruction Height Error < 5.5 cm (median 3.87 cm)

PID Benchmark Accuracy (%)

Query Type	TCPCN [8]	PCT-T MAML (sup.)	MAML (rand.)	MAML (pre-train)
Q1(same session)	96.18	98.99	76.67	94.44
Q2(cross-session)	35.10	32.05	54.44	70.00
Q3(cross-subject)	57.81	40.09	46.67	48.89

- Supervised:** PCT-T outperforms TCPCN $\rightarrow$ +2.81pp.
- Few-Shot:** Pre-trained MAML vs Random $\rightarrow$ +15.56pp.
- mmGait Benchmark:** PCT-T leads by +11.5pp on fixed trajectories [14].